

03: Inference for SLR

Stat 230 - Spring 2026

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library(tidyverse)
library(broom)
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1 Part 1: On paper¹

Biologists know that the leaves on plants tend to get smaller as temperatures rise, and average temperatures have been getting hotter over time. A sample of 252 leaves from the species *Dodonaea viscosa* subspecies *angustissima* (broadleaf hopbush) have been collected in a certain region of South Australia. Is there an association between leaf width and year? Below is the coefficient table from a simple linear regression model.

Note: The width is the average width, in mm, of leaves, taken at their widest points, that were collected in a given year.

- a. Report the fitted regression equation.

 Solution

$$\hat{y} = 37.723 - 0.018x$$

- b. Interpret the slope of the regression line in context of the data.

¹Adapted from *Stat 2: Modeling with Regression and ANOVA*

term	estimate	std.error	statistic	p.value
(Intercept)	37.723	8.575	4.399	<0.001
Year	-0.018	0.004	-4.029	<0.001

Solution

As year increases by 1 (i.e., from one year to the next), we expect the average leaf width of this species to decrease by 0.018 mm.

- c. Calculate a 95% confidence interval for the slope of the regression line. (Use the slides to find the appropriate critical value). Interpret this interval in context of the data.

Solution

To find the critical value, we need 2.5% on either side:

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qt(.025, df = 250)
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[1] -1.969498
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The 95% confidence interval for the slope is given by:

$$-0.018 \pm 1.97(0.004) = (-0.02588, -0.01012)$$

- d. What hypotheses are being tested with the statistic and associated p-value in the (Intercept) row of the table? What conclusion can you draw?

Solution

The summary output provided by R always conducts a two-tailed hypothesis test of whether a regression coefficient is equal to 0. In other words it tests

$$H_0 : \beta_0 = 0 \text{ vs. } H_a : \beta_0 \neq 0$$

Based on a test statistic of $t = 4.399$, $df = 250$, and a p-value < 0.001 , we find very strong evidence that the intercept is not zero. (This is outside the range of the observed data, so I won't interpret it in context.)